

**Interactive visual analysis of weather prediction data during an aircraft campaign for
weather research
(AI5253) Winter 2025/26**

Synoptic Analysis of a North Atlantic–European Cyclone

**Analysis (~Fri 12Z) – satellite and radar imagery and DWD/UKMO surface analyses; put Met.3D
visualizations into context**

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SLIDE 2: CASE OVERVIEW

INTRODUCTION

Case Overview

Analysis Time

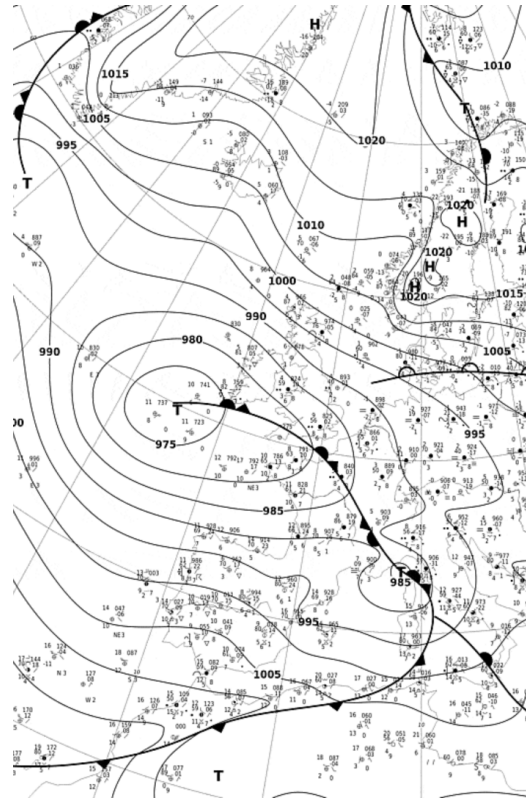
06 February 2026 – 12:00 UTC

Region of Interest

North Atlantic & Europe

Data Sources

- DWD Surface Analysis
- Radar Observations (DWD / KNMI)
- EUMETSAT Water Vapour Satellite
- ICON Model Visualized in Met.3D



This slide presents the general case study and data sets to be used in the analysis. The area of interest is the North Atlantic Ocean and the Western Europe where the mid-latitude cyclones are common and move eastward along the storm path.

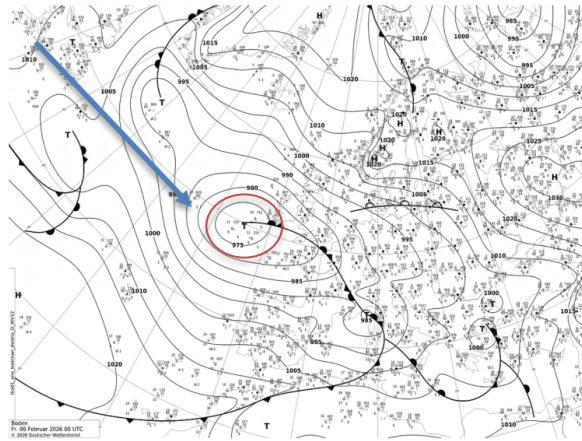
The time of analysis of 06 February, 2026, 12:00 UTC is in a stage where the cyclone is already developed. This enables us to see distinctly the synoptic pattern of the storm system.

A number of meteorological data were utilized in the analysis:

- DWD surface analysis, which gives the data regarding the distribution of pressure and the frontal boundaries.
- Radar images of DWD and KNMI that reveal the occurrence and strength of precipitation.
- EUMETSAT satellite imagery on water vapour, which is utilized in monitoring moisture distribution and upper-level atmospheric dynamics.
- Outputs of ICON numerical weather prediction model visualized through Met.3D software.

This is due to the fact that by using various datasets, a more accurate understanding of the structure of the cyclone can be made since the datasets focus on different facets of the atmosphere.

SLIDE 3: SYNOPTIC SITUATION AT THE SURFACE



SURFACE ANALYSIS

Synoptic Situation at the Surface

Deep Atlantic Low

~975 hPa low-pressure center located west of Europe, indicating a mature and intense cyclone at the synoptic scale

Frontal System

Well-developed frontal system present, with a cold front extending southeastward from the low-pressure center

Pressure Gradient

Strong pressure gradient around the cyclone, with a secondary weaker low identifiable over Central Europe

High-Pressure Ridge

A high-pressure ridge over Eastern Europe contrasts with the active low-pressure system dominating the western domain

The superficial examination shows the synoptic structure of the cyclone on the sea level. The North Atlantic has a deep low-pressure system having a central pressure of about 975 hPa that is west of Europe.

Very low pressures depict a very powerful and developed extratropical cyclone. The surface chart also indicates that a well-developed cold front is flowing south east of the cyclone centre. Cold fronts take place when a cold piece of air moves in front of warmer air, pushing it upwards.

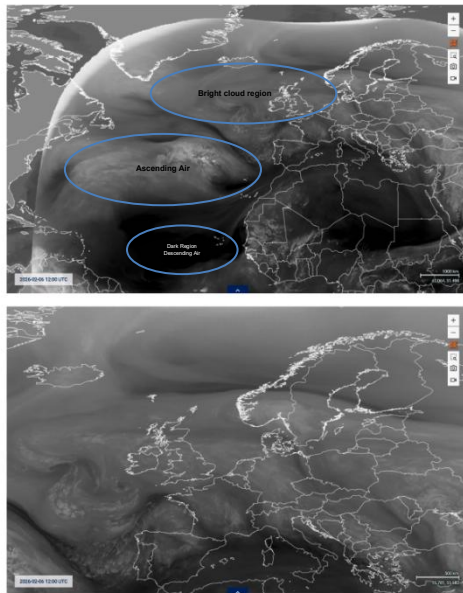
Cloud formation and precipitation occur due to the increasing move of the warm air along the frontal boundary. This process is likely to cause regular series of precipitation on the front. A close spacing of isobars also indicates that the surface map had a strong pressure gradient.

The stronger the gradient of the pressure the stronger the wind according to the meteorological theory. Also, there is a high-pressure area at the eastern side of Europe in comparison with the deep Atlantic low. The difference in the pressure leads to a complete cycle in the atmosphere and changes the movement of the cyclone.

SLIDE 4: SATELLITE ANALYSIS

SATELLITE

Satellite Analysis – 06 February 2026, 12 UTC



Water Vapour Imagery

The satellite image at 12 UTC shows a well-developed spiral cloud structure over the North Atlantic, confirming the mature stage of the cyclone.

Key Features

- Pronounced cloud band extending southeastward along the frontal zone
- Contrast between bright and darker regions reflects differences in moisture content
- Dry intrusion wrapping in from the southwest — characteristic of dynamically active mid-latitude cyclones
- Strong interaction between upper-level and surface processes confirmed

The satellite imagery gives valuable data on the moisture in the atmosphere and the cloud formation. The vapour satellite images of water depict an enormous spiral shape of clouds, which is typical of full-grown extratropical cyclones.

The dry intrusion is one characteristic that can be spotted in the satellite imagery, that is, it looks like a darker area that is enveloping the cyclone circulation to the southwest. Dry intrusion This incursion takes place when dry air at the top of the troposphere falls into the cyclone.

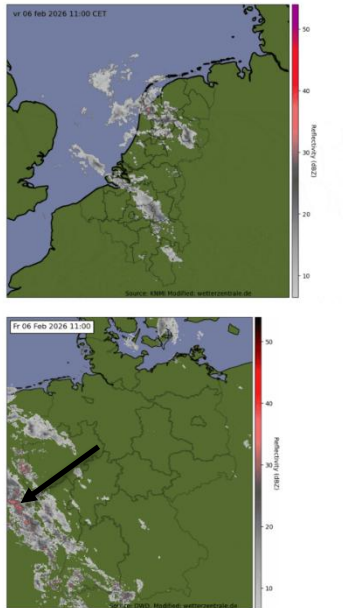
This process is very commonly linked to vigorous dynamical activity and can make storm systems severe.

The satellite image also indicates that there is a long band of the cloud that points to the frontal zone that is the cold front as illustrated in the surface analysis.

All these characteristics ensure that the cyclone was dynamically active and mature, at the moment of observation.

SLIDE 5: RADAR OBSERVATIONS

RADAR
Radar Observations – Frontal Precipitation Structure



Precipitation Band Structure

Radar observations at 12 UTC reveal a well-defined precipitation band aligned SW-NE across parts of Western and Central Europe, corresponding to the cold frontal zone.

Key Observations

- Strongest reflectivity values concentrated along the frontal boundary
- Embedded areas of higher intensity suggest locally stronger convective elements
- Precipitation primarily organized along the frontal zone
- Consistent with both surface analysis and satellite frontal position

Weather radar offers a high-level of observation of precipitation. According to the radar images, there is a precipitation band that spans south west to northeast through Western and Central Europe.

This band is similar to the point of the cold front found in the surface analysis. On the frontal boundary, cold air moves towards the front and lifts warm moist air, which gives rise to clouds and precipitation.

The radar images further demonstrate the local occurrences of greater reflectivity, which reflect greater areas of more precipitation. These can be the small-scale convective features embedded in the bigger frontal set-up.

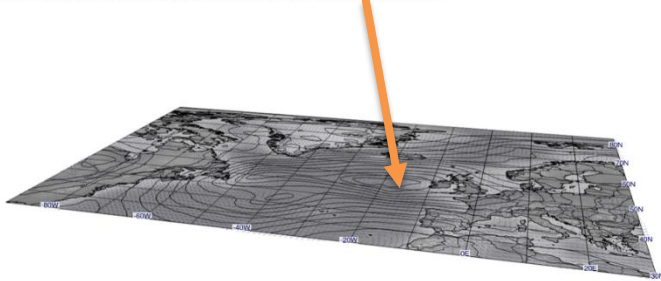
The radar images affirm the arrangement of the precipitation system and give a specific insight on the distribution and intensity of the rainfall related to the cyclone.

SLIDE 6: Met.3D MEAN SEA LEVEL PRESSURE

MET.3D

Met.3D – Mean Sea Level Pressure

1u 2026-02-05T12:00:00 (-120 hours from Tue 2026-02-10T12:00:00)



View 1 (Scene 1)

The Met.3D visualization reproduces the main synoptic features observed in the DWD surface analysis with good agreement in position and intensity.

Model Fidelity

- Deep Atlantic low clearly represented at similar position and intensity
- Pressure gradient well defined, especially south of the low center
- Tightly packed isobars indicate strong surface winds
- Frontal structure inferred from pressure field consistent with observational analysis

Met.3D is a three-dimensional visualization program that analyzes the output of the numerical weather prediction models. In this work, ICON model was represented in Met.3D that was used to investigate the pressure structure of the cyclone.

The principal characteristics observed on the observational surface analysis are reproduced by the model output. It is evident that there is a deep Atlantic low at a position that is almost similar and the same intensity.

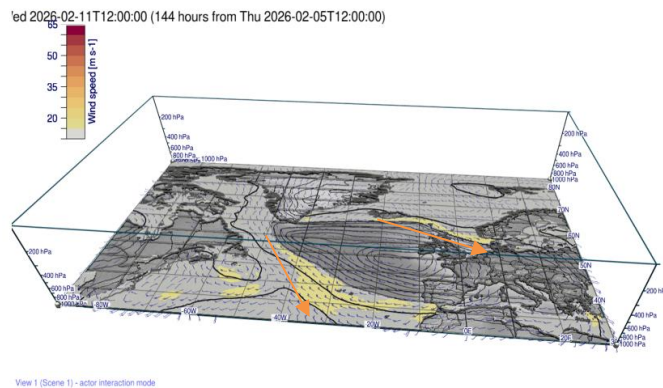
The visualization shows the isobars being tightly clustered around the cyclone center and this depicts that there are strong pressure gradients. Such pressure gradients have direct relationships to development of strong surface winds.

The ICON model simulation -observational data agreement indicates that the model is able to capture the large-scale structure of the cyclone.

SLIDE 7: Met.3D WIND FIELD AND PRECIPITATION

MET.3D

Met.3D – Surface Wind Field & Precipitation



Wind Distribution

Strongest winds south and southwest of the cyclone center, where the pressure gradient is most pronounced. Wind field follows isobars with clear cyclonic circulation.

Geostrophic Adjustment

Consistent with geostrophic theory – tighter pressure gradients produce stronger winds, confirming the physical relationship between pressure gradient and surface wind intensity.

Precipitation Structure

Well-defined frontal precipitation band associated with the Atlantic cyclone. Strongest areas coincide with regions of enhanced frontal lifting.

Model vs. Observations

Model reproduces the overall structure and position of the precipitation band well. Minor differences in small-scale intensity expected due to model resolution effects.

The wind field and structure of the precipitation of the cyclone can also be studied using the Met.3D visualization.

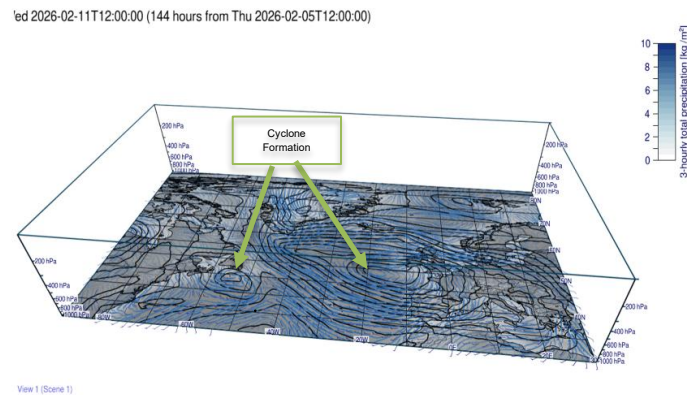
The greatest winds are south and southwest of the center of the cyclone in which there is the greatest pressure gradient. In such areas, the winds follow the isobars and take a cyclonic ring pattern round the low-pressure center.

This correlation is in line with geostrophic balance where Coriolis force and pressure gradient force are equalized to form large-scale atmospheric flow.

Precipitation field exhibits a distinct band which coincides with the frontal line. Frontal lifting forces heat hot air which then rises and results in the most vigorous precipitation in the areas where they occur.

The model can reproduce the total structure of precipitation, but there are some minor variations in the strength of precipitation because the model does not have the capability of resolving the model.

SLIDE 8: DISCUSSION AND MODEL EVALUATION



DISCUSSION

Discussion & Model Evaluation

Observations vs. Model

Strong consistency between DWD surface analysis, EUMETSAT satellite imagery, KNMI/DWD radar, and ICON model output across all analysed fields

Satellite Insight

Dry intrusion and spiral cloud structure confirm dynamically active cyclone stage, supporting the surface-based frontal analysis

Radar Confirmation

Frontal precipitation band structure in radar aligns well with model-predicted precipitation, validating ICON's frontal positioning

Model Limitations

Minor discrepancies in small-scale precipitation intensity are attributable to parameterization effects and resolution constraints inherent to the ICON model

The argument is based on comparing the ability of the numerical model to replicate the observed atmospheric conditions.

Surfaces analysis, satellite images, radar observations as well as icons model output are all compared to reveal a lot of consistency between the datasets. According to all sources, there is the existence of a well-developed cyclone with an explicit frontal system and well-organized precipitation structures.

The satellite pictures prove the presence of the spiral cloud formation and dry intrusion in the cyclone. Radar images depict that there exist precipitation bands going along the modeled frontal area.

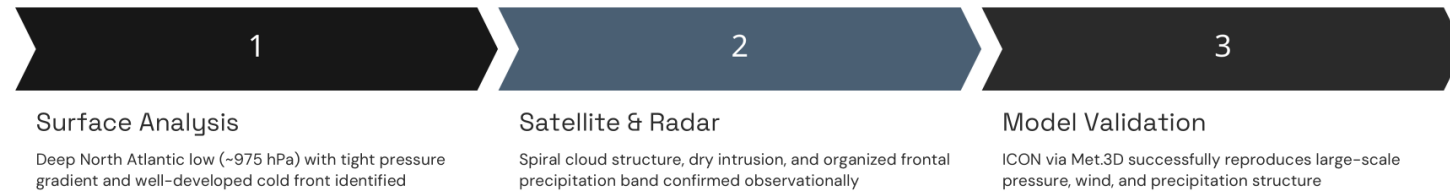
There are small differences that arise in the intensity of small-scale precipitation which can be expected since the numerical models parameterize some processes in the atmosphere.

In general, the model can be considered a good representation of the big-scale construction of the cyclone.

CONCLUSION

CONCLUSION

Summary & Conclusions



Key Takeaways

Strong consistency between observations and ICON model representation at 12 UTC on 06 February 2026

Satellite imagery and radar confirm the mature cyclone stage and dynamically active frontal system

Met.3D demonstrates reliable simulation of synoptic-scale features including pressure, winds, and precipitation

Thank You

Questions & Discussion

The case study shows that a combination of several observational datasets and model simulations can be used to study the structure of a synoptic-scale cyclone. The surface analysis revealed a deep low-pressure area of about 975 hPa which had a developed cold front. The presence of the mature cyclone was verified by satellite data with the help of spiral shape clouds and dry intrusions.

Radar scans indicated that there were arranged precipitation bands parallel to the frontal boundary. The ICON model visualization with Met.3D was effective in the reproduction of the key atmospheric features such as patterns of pressure and wind circulation and structures of precipitation.

Altogether, the obtained outcomes indicate high consistency between observational and model simulation data, which explains the success of the meteorological observations and sophisticated visualization tools.

LEARNING ABOUT THE UNKNOWN APPLICATION AREA

The given project also gave me an opportunity to work in an area of application that I initially was not familiar with, i.e. synoptic meteorology and atmospheric analysis. Being a student whose background was mainly in software development and data analysis, the process of working with the meteorological data meant that I needed to acquire other scientific principles of pressure gradients, cyclone dynamics, and frontal systems. The case study of a real atmospheric case study enabled me to see how complicated weather systems form and as well as how scientists interpret atmospheric data. This experience has shown that the computational tools and visualization techniques could be used in assisting the scientific research in the fields that are not my major area of interest.

LEARNING ABOUT VISUAL ANALYSIS

Learning to analyze the visual aspect of data to understand the complex data was one of the most significant parts of the project. Atmospheric processes constitute big and multidimensional data that cannot be interpreted through numerical numbers only. Patterns that are easier to identify with the help of visualization tools include:

- Pressure gradients
- Frontal boundaries
- Wind circulation patterns
- Precipitation structures

A combination of several visual datasets enables the meteorologists to know how various processes in the atmosphere come together in a storm system.

EXPERIENCE WITH MET.3D

In this project, I got to work with Met.3D, as this was a new tool to me. I have used the Met.3D software to come up with the visualizations that were displayed in the presentation depending on the topic of the project.

It was also interesting and challenging to learn how to operate Met.3D. The software enables one to explore atmospheric data in three-dimensional format and this is more convenient in understanding the vertical and horizontal makeup of weather systems.

Using the tool assisted me in gaining a better insight into how pressure systems relate with wind patterns and precipitation. In general, the process of learning Met.3D was a good experience and it was pleasant to work with this new technology.

PRESENTATION AND DISCUSSION EXPERIENCE

The presentation took around 10 minutes and the questions and discussion took up around 5 minutes. In the discussion session, a number of classmates posed questions and gave suggestions regarding the analysis of the cyclone and the visualizations that were employed in the presentation. I could respond to their questions and explain the ideas that I discussed in my slides. The presentation aided in solidifying my knowledge on the subject and made the audience have no confusion as to what I was analysing.

The questions and feedback provided by the audience were positive and they assisted in making the presentation more comprehensible.

USE OF AI TOOLS

The primary reason why AI tools were employed when preparing this project was to facilitate the organization of the content of the presentation and paragraph organization. The first structure of the PowerPoint slides was also generated with some assistance of AI. Nevertheless, the meteorological data was scientifically interpreted, and the cyclone analysis was done and the Met.3D visualizations developed, as an independent project work. There was the use of AI tools as the writing helping tools to enhance the clarity and the organization.

FEEDBACK ON MET.3D

Met.3D is an effective visualization software that enables the user to view the information regarding the atmosphere in a three-dimensional format. It is one of its primary benefits because it allows one to integrate various atmospheric variables into one visualization space. This facilitates effortless comprehension of the space associations among pressure systems, wind patterns and precipitation. The software may however initially be hard to master by new users. More tutorials or documentation may assist beginners to familiarize themselves with the interface faster. After having the fundamentals of the functions, Met.3D can prove to be a very handy tool in visualizing the processes of the atmosphere.